

Real-Time Carbon Footprint Monitoring System

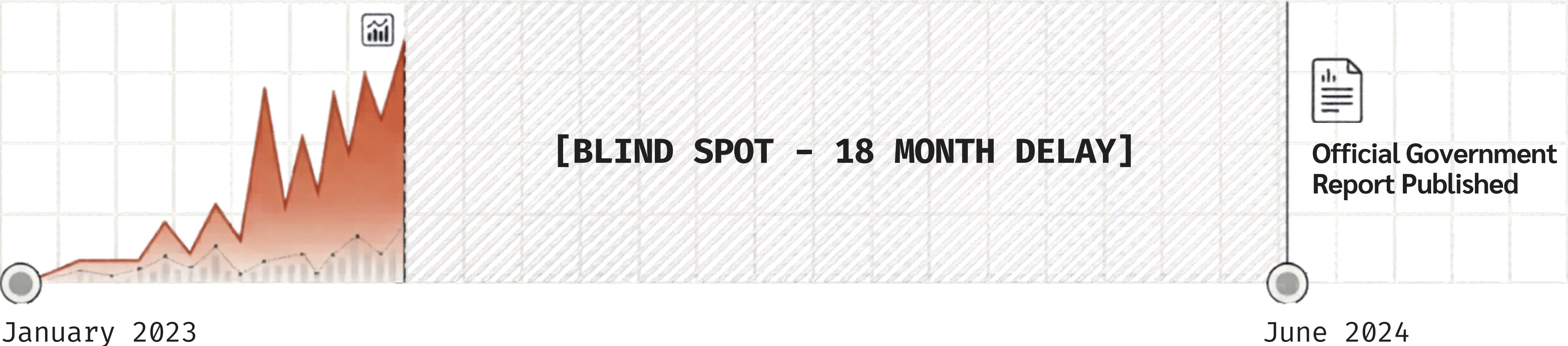
A Proactive Intervention Architecture for the
Climate Crisis (UN SDG 13) using Big Data
and Machine Learning

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[PROGRAM]:
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The Climate Crisis Won't Wait, Yet Data Is 18 Months Behind



Global Crisis



In 2023, global CO2 emissions hit a record 36.8 billion tons. The Paris Agreement requires halving this rate by 2030.



Collapse of Current System



Traditional systems report emissions annually. If a new industrial zone creates pollution, policymakers only learn about it 12-18 months later.



Solution Vision



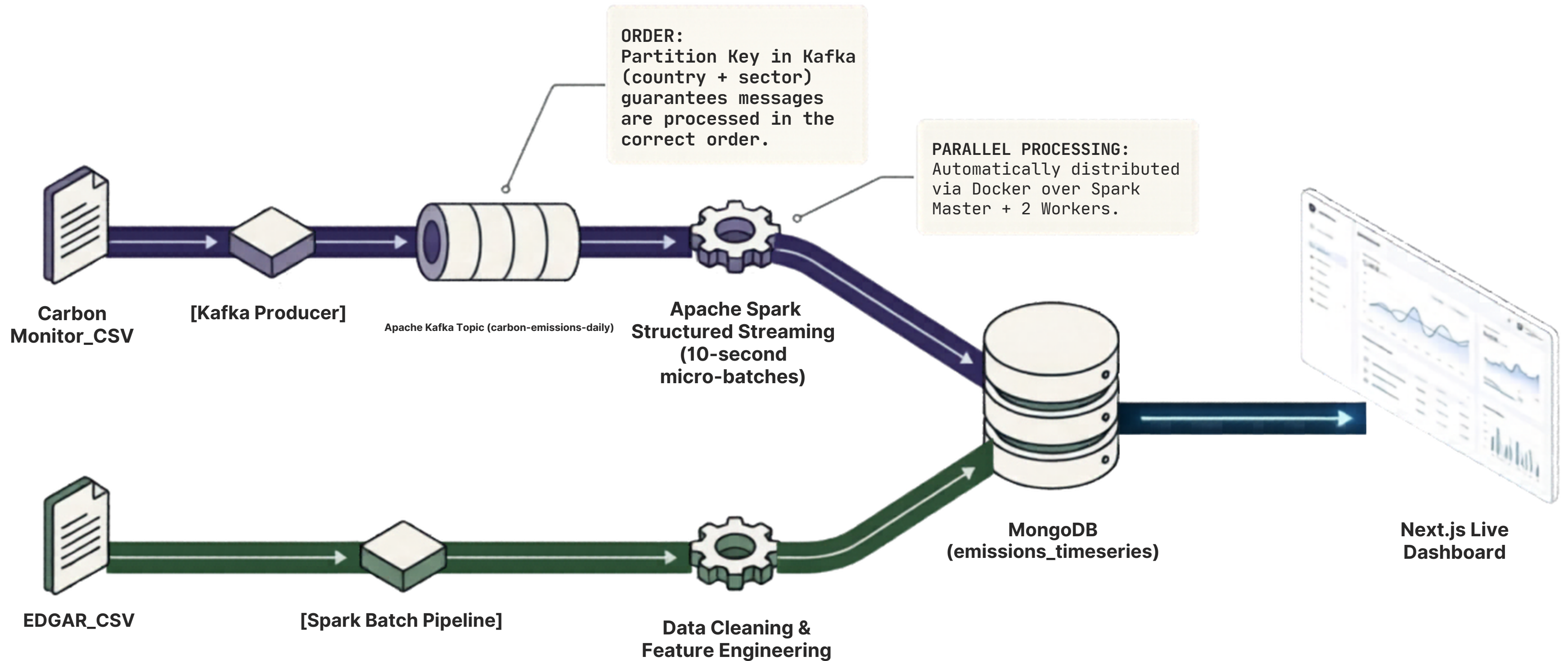
To achieve UN SDG 13 goals, a platform for daily and sector-based estimation was built using distributed computing and machine learning.



The Foundation from Past to Near Real Time

Data Set	Role in Project	Scope and Size	Systemic Constraints
EDGAR (European Commission)	[BATCH LAYER] Historical foundation. Spark Batch processing and Spark MLlib training.	1970-Present, 200+ Countries (50 GB raw data).	Only annual resolution. Cannot be used for real-time monitoring.
Carbon Monitor (Nature Sci. Data)	[SPEED LAYER] Real-time flow. Kafka streaming, backbone of LSTM and its its training.	Daily CO ₂ estimates. 10 Countries × 6 Sectors × 5 Years (109,200 records).	Not direct measurement, estimate from electricity/traffic logs (±%5-10 uncertainty).
Kaggle Individual Footprint	[CLASSIFICATION] Only for personal footprint sub-task. Not confused with macro data.	5,000 individual lifestyle records.	Based on self-reporting, does not have national inventory quality.

Lambda Architecture: Synchronized Flow of Data from Two Streams



Critical Architectural Decisions: Why This Tech Stack?



KAFKA ARCHITECTURE

Traditional Method: Reading line by line from file (Pandas).

Our Choice: Why Apache Kafka?

If the consumer slows down, it buffers messages. No data loss. Multiple models can pull data from the same topic simultaneously.



SPARK DISTRIBUTED PROCESSING

Traditional Method: In-memory processing on a single machine.

Our Choice: Why Apache Spark?

50 GB of EDGAR data cannot be processed with Pandas. Spark shards the data and automatically distributes it across multiple machines by scaling horizontally. Workers can be added without changing the code.



MONGODB NOSQL

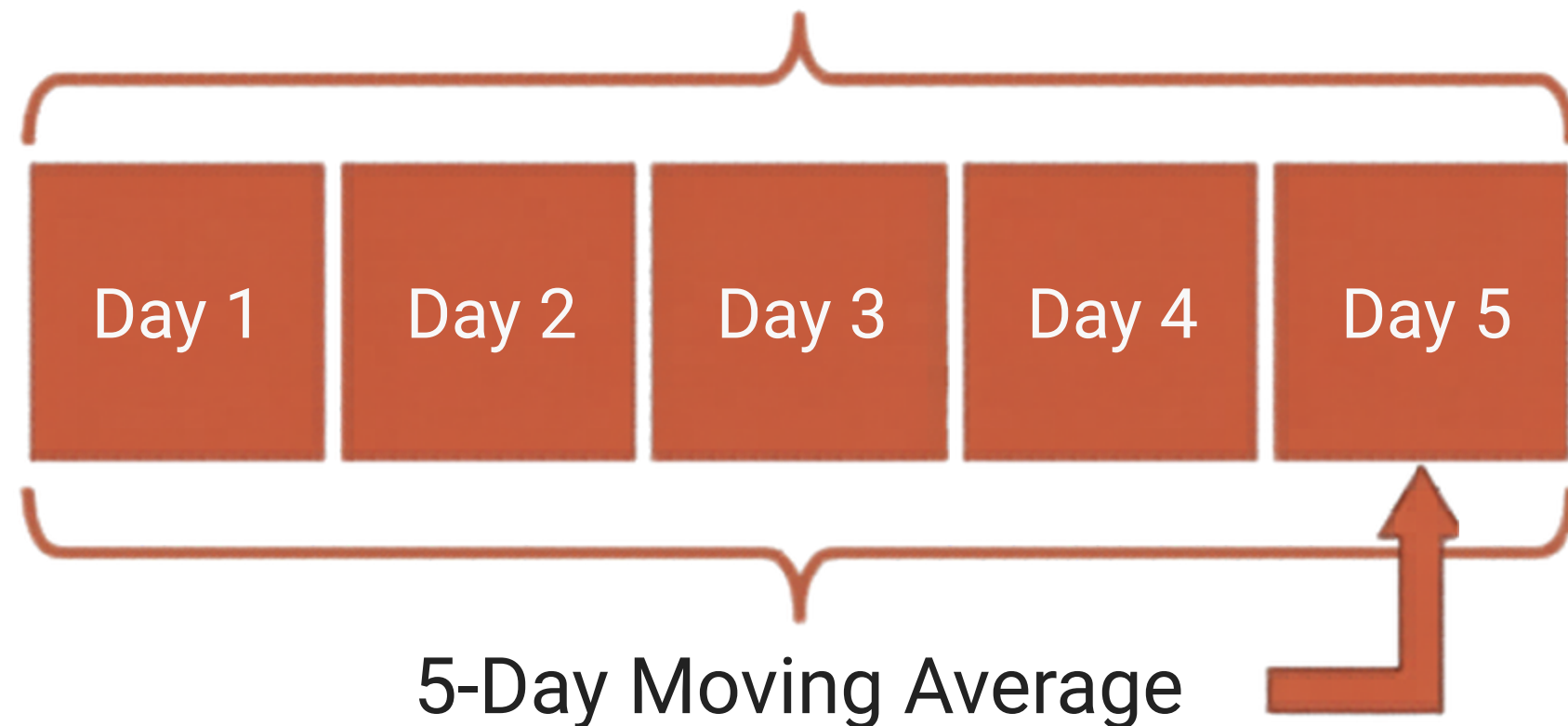
Traditional Method: PostgreSQL (Relational tables)

Our Choice: Why MongoDB?

Different metadata fields for countries are stored flexibly with NoSQL (no need for complex JOIN operations). It also has native Time-Series collections for timestamps data.

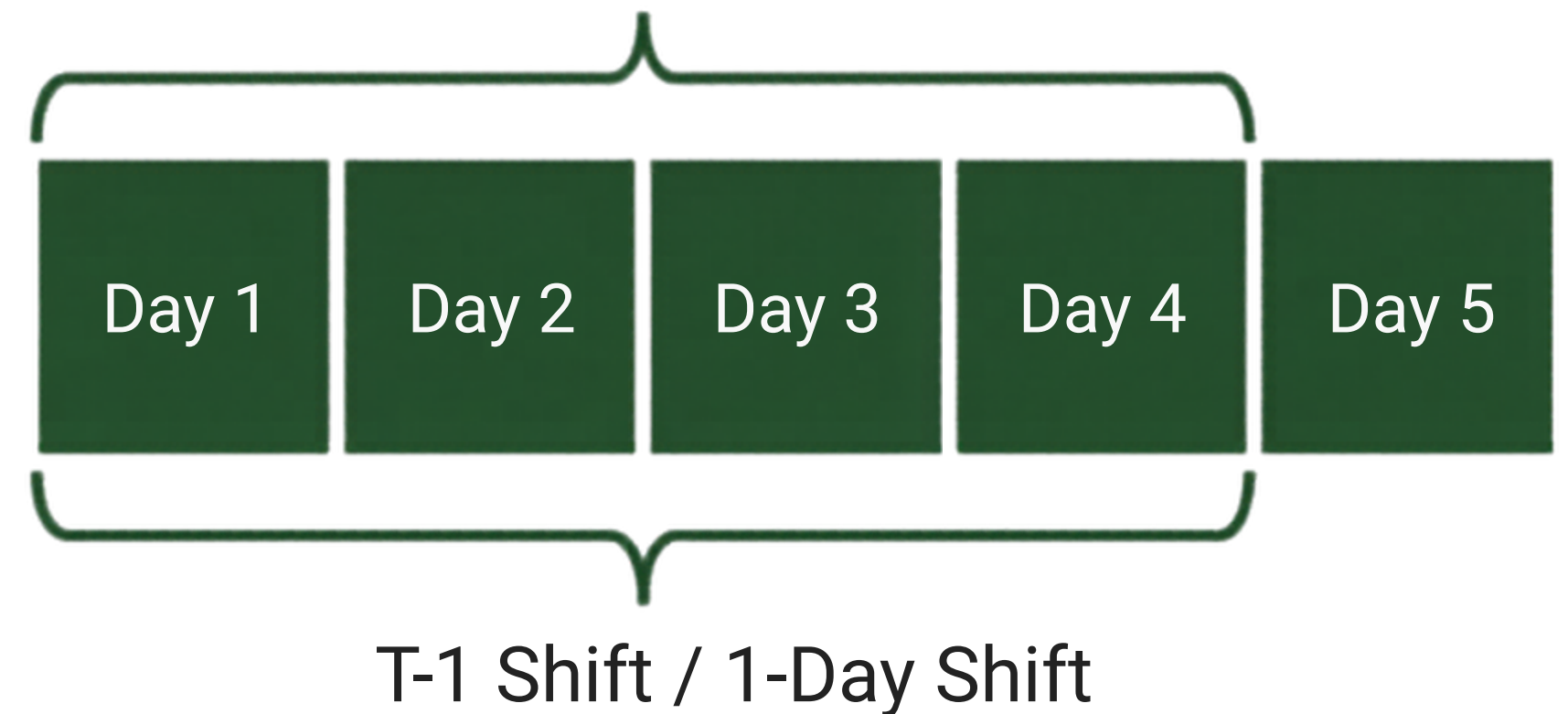
Isolation of Target Leakage in Time Series

Flawed Model



DIAGNOSIS: The model is “cheating”
by learning today’s data.
(Target Leakage)

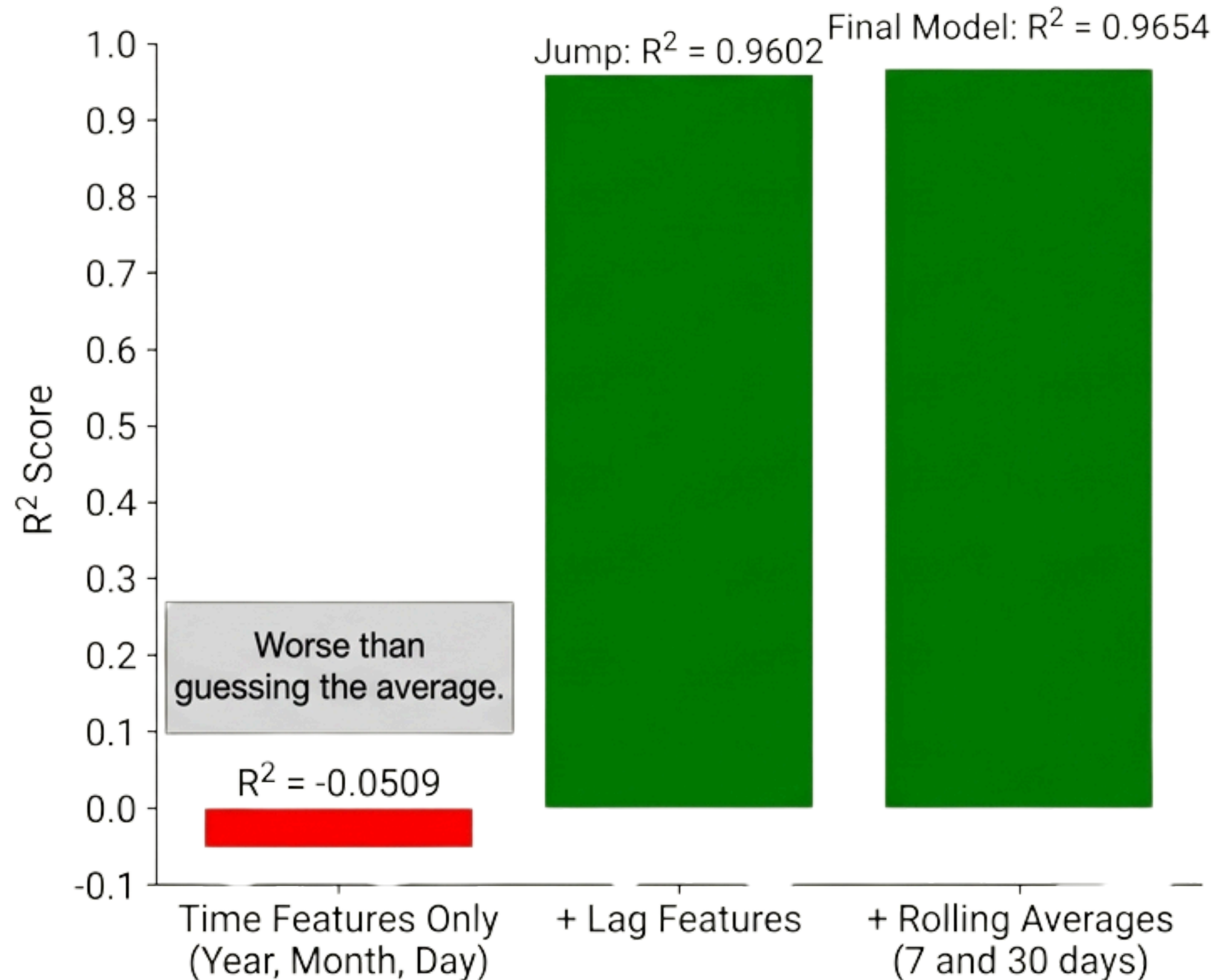
Validated Model



SOLUTION: Data was shifted by 1 day and
included in moving averages. The model
was forced to make honest predictions.

While training the XGBoost model (with 15 features), it was discovered that Pandas’ default moving average calculation included “that day’s” (target) data. A simple “shift” operation prevented the model from seeing the future. Train and test sets were split sequentially along the time axis (80%/20%) without shuffling, completely sealing the leakage.

Ablation Study: What Actually Drives the Model?



SYSTEM_INSTIGHT: The model learns nothing from calendar or seasonality alone.

The main predictive power comes from the **autocorrelation of emission dynamics** (especially **lag_7d** = 65% importance).

XGBoost; captures these dynamics without falling into overfitting, using early stopping (179/500 trees).

Model Performance and Diagnostic Matrix

Model	Performance Metrics	Status and Architectural Limits
Baseline (Persistence / Yesterday = Today)	$R^2 = 0.9915$ MAE = 0.0216	Very high autocorrelation. However, it cannot capture structural breaks like COVID-19 and does not offer SHAP explainability.
XGBoost Regression	$R^2 = 0.9654$ MAE = 0.0402 RMSE = 0.1098	Performs simultaneous generalization across 10 countries and 6 sectors, understands lockdowns. Trains in 12 seconds on CPU.
LSTM (PyTorch - 30-day window)	$R^2 = 0.8533$ MAE = 0.1000 RMSE = 0.1231	Unlike ARIMA, it solves non-linear seasonality without assumptions (2 layers, 128->64).
Spark MLlib Random Forest	$R^2 = 0.6644$ MAE = 104.57 RMSE = 237.27	Margin of error stems from the annual EDGAR data scale (MtCO ₂ /year). Low R^2 reflects Docker cluster resource limits (Java OOM prevention).

Sectoral Volatility and Error (MAE) Analysis

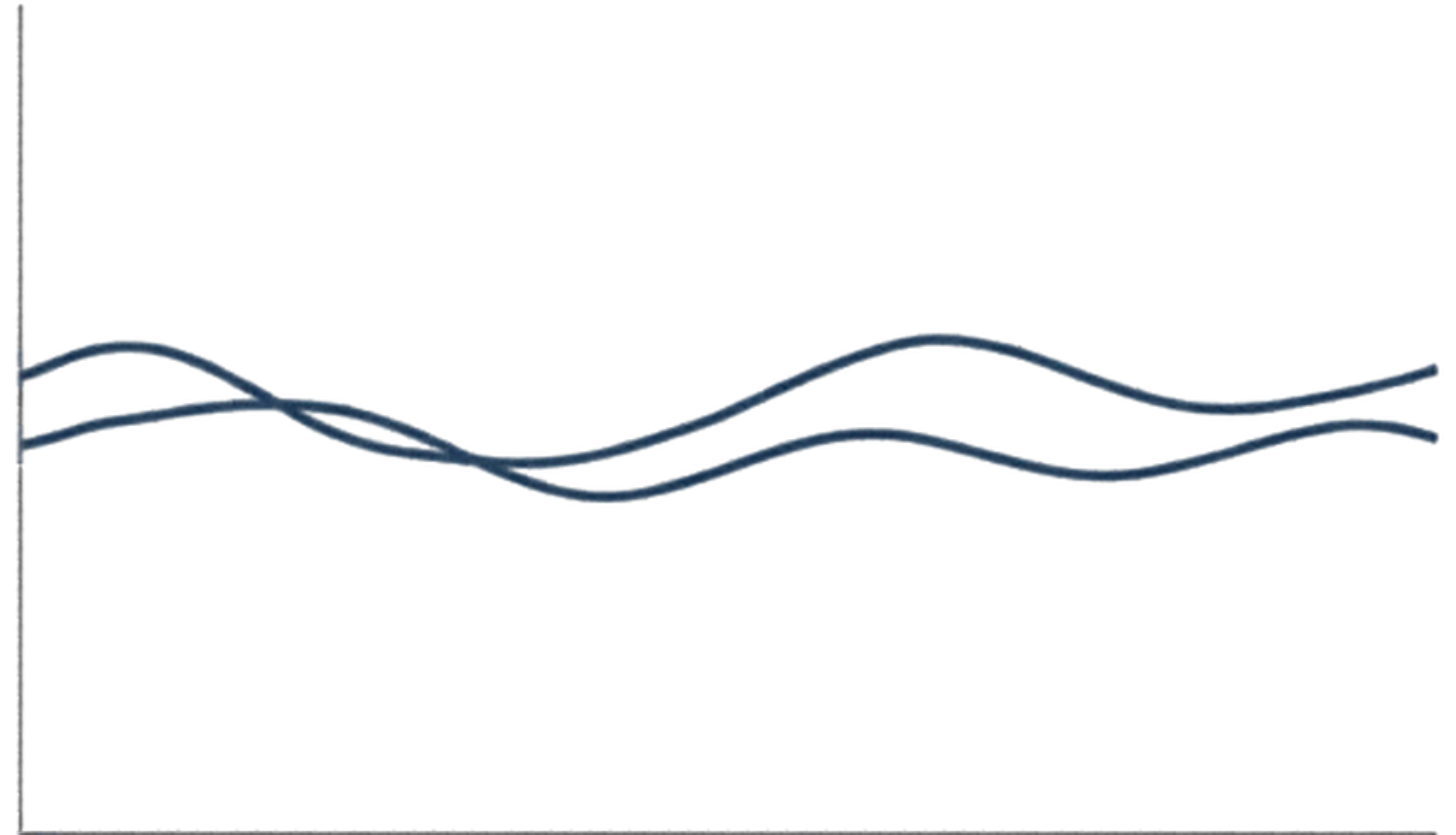
Ground Transport – High Volatility



MAE = 0.131 MtCO₂/day (Highest error)

Directly linked to daily human behavior. The most difficult sector to predict due to weekend drops and holidays.

Maritime & Aviation – Low Volatility

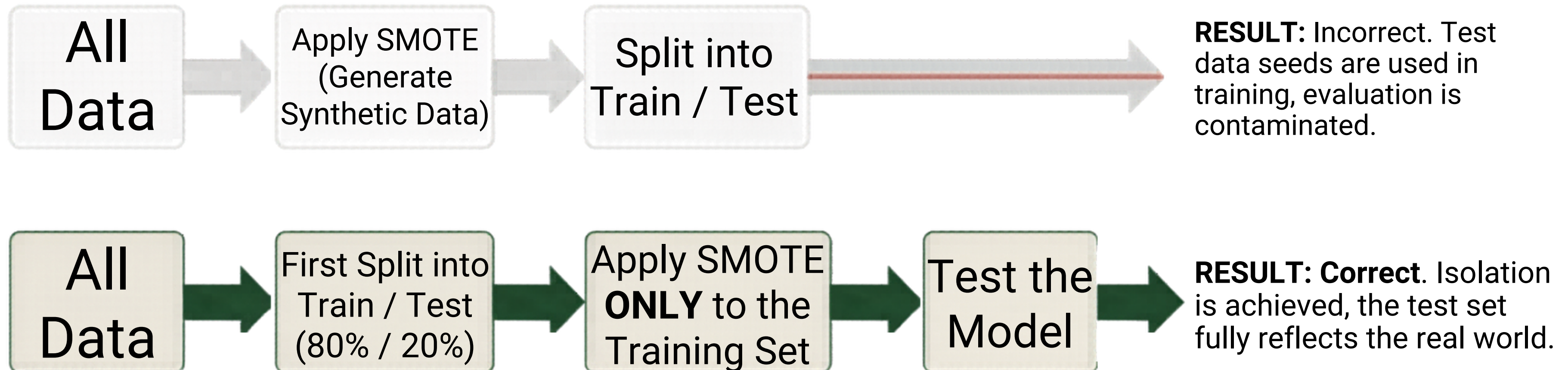


Maritime MAE = 0.0036 | Aviation MAE = 0.0052

These are sectors with small-volume, stable, programmed, and consistent patterns. Machine learning models capture these patterns with almost zero error.

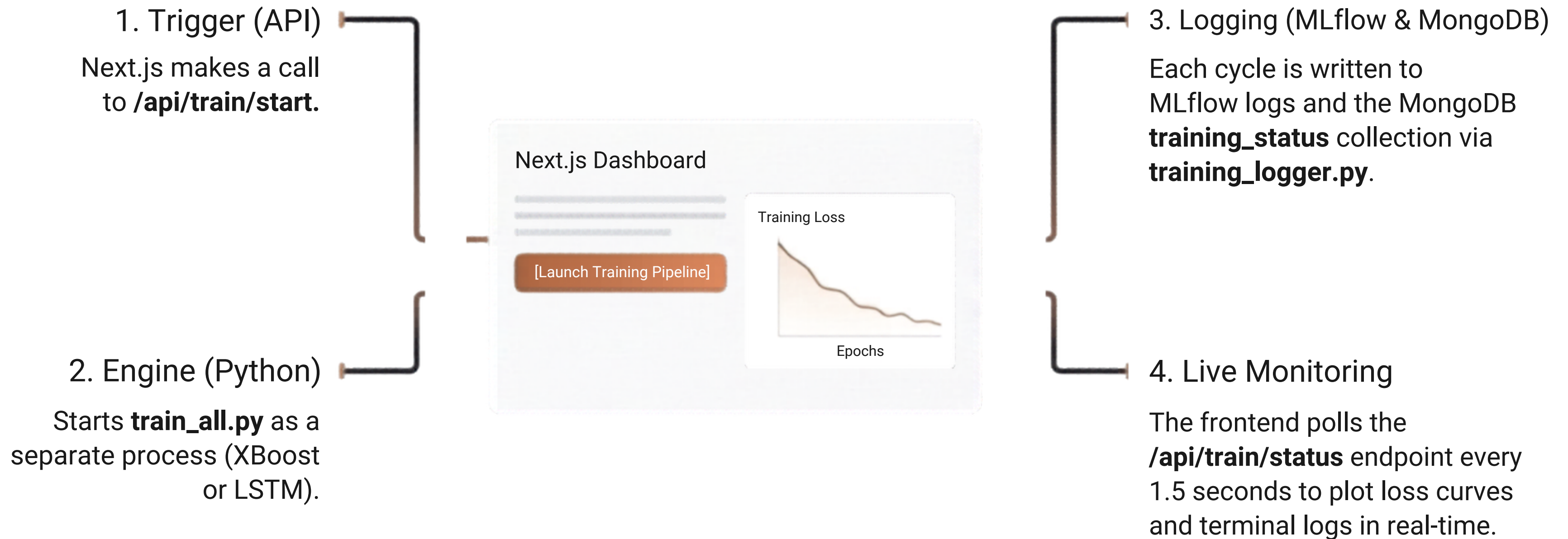
Isolation of the SMOTE Layer in Class Imbalance

Individual carbon footprint classification (Kaggle 5,000 records): Low (9.9%), Medium (73.6%), High (16.5%)



Model: XGBoost Classification | Accuracy = 85.4% | F1-Score = 85.55% | Precision = 85.89% | Most decisive feature: 32.9% Transportation Type

Synthesis: Transformation of the Complex Backend into a Live Command Center



The big data architecture is made completely invisible and seamless for the end-user. The system not only visualizes data but also enables live model training via the browser.

Strategic Synthesis: System SWOT Analysis

Strengths

End-to-end architecture: Full integration from Raw_CSV to live dashboard.

Production quality: Exponential backoff in Kafka, early stopping, one-click Docker deployment.

SHAP-supported explainability and 96.5% variance explanation with XGBoost.

Weaknesses

- **Kafka stream operates with a CSV simulation instead of a real live API.**
- **Spark MLib** has not been benchmarked on a real cloud infrastructure (AWS EMR).

Opportunities

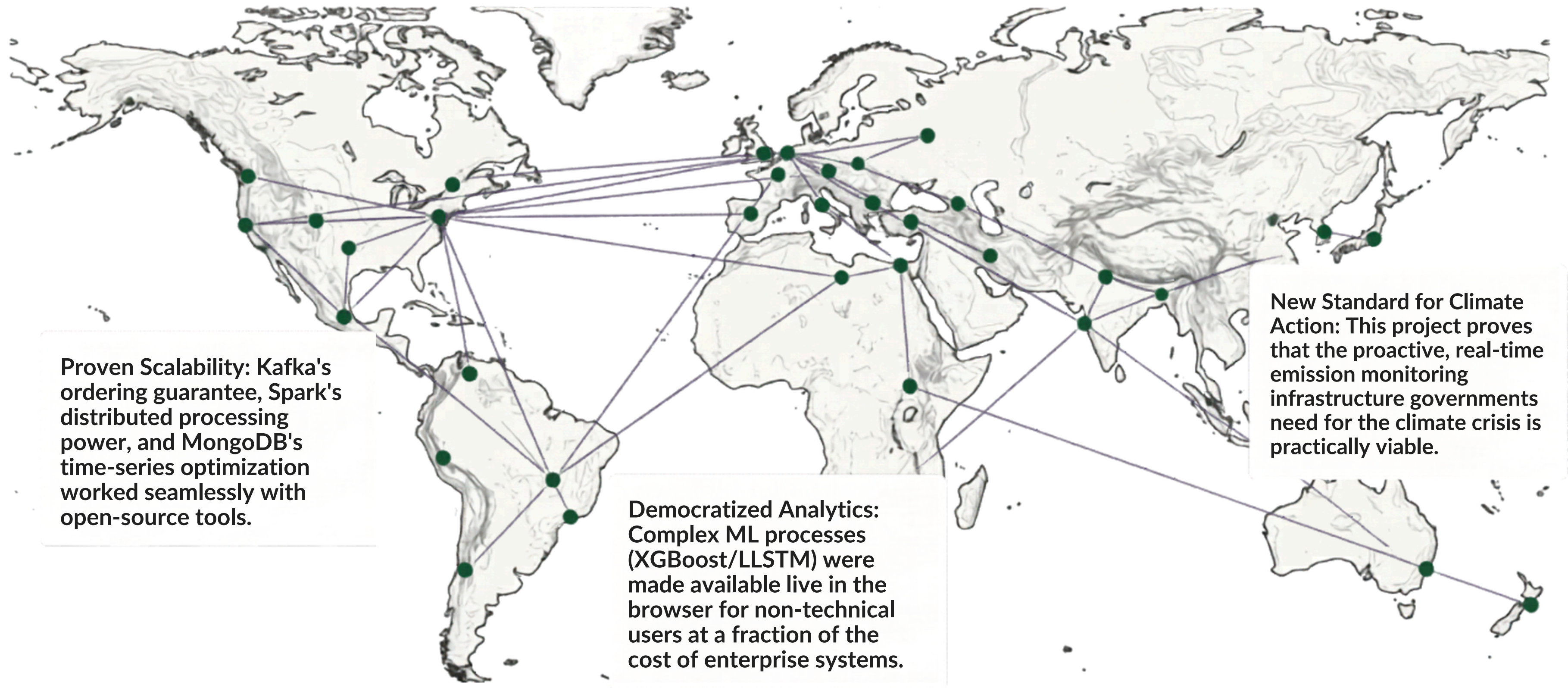
SaaS vision: Scaling by taking into production with AWS ECS + MSGM + Atlas.

Potential to enable governments to respond to emission shocks within 24-48 hours instead of 18 months.

Threats

- The $\pm 5\text{-}10\%$ uncertainty of Carbon Monitor data limits the ceiling prediction power.
- Risk of structural breaks, like the COVID period, creating model drift.

Conclusion: Transition from Waiting to Action (UN SDG 13)



THANK YOU